

Attacking the Lucas Conjecture and Looking at Frege vs. Extended Frege:

What I Proved, What I Could Not, and How Hard These Problems Are

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Abstract

I worked by hand, using computation once to test a proof strategy. Results:

- **A reformulation.** The Lucas conjecture is equivalent to a clean statement about *variable-window Fibonacci recurrences*.
- **A partial theorem.** The golden speed limit $\varphi^{1/3}$ is *exactly* right for every library in which each result cites at most two others. The proof is complete and short.
- **The obstruction.** The general case fails at one identifiable phenomenon, *runs of near-equal values*, and I explain why every simple potential argument breaks there.

I did not prove the full conjecture. I did not, and cannot honestly claim to, make progress on Frege vs. Extended Frege. That is a famous problem at the frontier of complexity theory, and I explain where it sits. I also say plainly how famous and how hard each problem is.

1 Are these real open problems? How hard are they?

The Lucas conjecture. It is *our* conjecture: we posed it two days ago.

- **Status.** It is “open” only in the sense that it is unproved. A literature search found no prior statement of it, though a specialist should double-check. It is not a recognized problem that anyone else is working on.
- **Difficulty.** My honest estimate is a solid research problem in extremal combinatorics. A clean proof would make a good paper in a combinatorics journal, especially with the exact Lucas-number answer.
- **Prize level.** It is nowhere near Fields-Medal level. It could fall to one good idea. It could also be subtler than it looks: Section 4 shows that the obvious methods provably fail.

Frege vs. Extended Frege. This is a genuinely famous open problem.

- **History.** It was posed in Cook and Reckhow’s founding paper of proof complexity (1979) and is still open after 45 years.
- **Even the first step is open.** Nobody can prove a super-polynomial lower bound for *Frege itself* on any tautology, and separating Extended Frege from Frege would require that.
- **Its circuit analogue is just as stuck.** The analogue is showing that circuits (computations with shared, *named* subresults) are super-polynomially stronger than formulas (no sharing). The best formula lower bounds for explicit functions are about n^3 (Håstad; Tal), and super-polynomial ones are a celebrated open problem.

- **Prize level.** A resolution would be a landmark in theoretical computer science, in the tier of results honored by the Gödel Prize or the IMU Abacus Medal. It is not a Millennium Prize problem, but it sits in the research program around P vs. NP. Whether it would be Fields-level depends on the mathematics a proof brought with it; I won't pretend to calibrate that.
- **My position.** I cannot prove it, and you should be very skeptical of anyone, human or AI, who claims a quick proof.

One clarification. “Prove Extended Frege” is not quite a thing. Extended Frege is a proof system, not a statement. The open question is a *separation*: are there tautologies with short Extended Frege proofs but no short Frege proofs?

2 The Lucas conjecture, reformulated

Recall. Results are added one at a time. Result t cites a set of d_t distinct earlier results and has unfolded size $u_t = 1 + \sum_{\text{cited } j} u_j$, the number of citation paths starting at t . The cost is $C = \sum_t (1 + d_t)$.

Conjecture 1 (Lucas). *At cost $3m$ ($m \geq 4$) the largest unfolded size is $2L_m - 2$. In particular, $u \leq K\varphi^{C/3}$ for a constant K .*

Lemma 1 (Greedy dominance). *Fix the sequence of citation counts $d_1, d_2, \dots$. Among all libraries with these counts, the largest values are achieved when every result cites the d_t largest values available.*

Proof. Compare any library with the greedy one, step by step. Keep the multiset of values of each, sorted in decreasing order.

Invariant: the greedy multiset dominates the other elementwise.

Step: if multiset A dominates multiset B , then the sum of the d largest elements of A is at least the sum of any d elements of B . So the greedy library's new value a is at least the other library's new value b . Inserting a into A and b into B , with $a \geq b$, preserves elementwise dominance.

Costs are identical, since they depend only on the d_t . □

The window form. In a greedy library, every non-leaf result cites the current largest value. Its own value is therefore larger than everything so far, so non-leaf values appear in *increasing* order $z_1 < z_2 < \dots$. A non-leaf step with window $d \geq 1$ is then

$$z_{\text{new}} = 1 + z_n + z_{n-1} + \dots + z_{n-d+1},$$

where leaves (value 1) fill in when the window reaches below the non-leaf values. The conjecture becomes:

In any variable-window Fibonacci recurrence, where each step chooses its own window d at price $d + 1$, the growth per unit price is at most $\varphi^{1/3}$.

Ordinary Fibonacci is window 2 at price 3, growing by φ per step: exactly $\varphi^{1/3}$ per unit. The d -bonacci recurrences grow by φ_d per price $d + 1$, and $\varphi_d^{1/(d+1)}$ is largest at $d = 2$:

window d	1	2	3	4	5	large
growth per step φ_d	1	1.618	1.839	1.928	1.966	$\rightarrow 2$
per unit price $\varphi_d^{1/(d+1)}$	1	1.174	1.165	1.140	1.119	$\rightarrow 1$

So among *constant* windows, 2 wins. The conjecture says that *varying* the window cannot beat it either.

3 A partial theorem: at most two citations

Theorem 2 (Golden speed limit for binary libraries). *If every result cites at most two others, then every unfolded size satisfies*

$$u < 7\varphi^{C/3}.$$

The Fibonacci library achieves $\Theta(\varphi^{C/3})$, so the rate $\varphi^{1/3}$ is exact for this class.

Proof. By Lemma 1 we may assume the library is greedy. The lemma preserves each $d_t \leq 2$, hence the class.

Potential. Let $\alpha = \varphi^{1/3}$, so $\alpha^3 = \varphi$. Let $z \geq z' \geq 0$ be the two largest values so far ($z' = 0$ if there is only one). Define

$$\Phi = \max\left(z + \frac{z'}{\varphi}, \alpha z\right).$$

The first piece is the classical Fibonacci invariant: it is multiplied by exactly φ per Fibonacci step. The second piece accounts for a “copy” step, which is cheap and almost doubles the second value. We show that each step of price p satisfies $\Phi_{\text{new}} + 7 \leq \alpha^p(\Phi + 7)$.

Leaf ($p = 1$). The top two can only change by z' rising from 0 to 1, so $\Phi_{\text{new}} \leq \Phi + 1/\varphi$. If the library was empty, then Φ goes from 0 to α , and $\alpha + 7 \leq 7\alpha$. Otherwise $(\alpha - 1)(\Phi + 7) \geq 7(\alpha - 1) > 1.2 > 1/\varphi$.

Copy ($d = 1, p = 2$). The new top two are $(z + 1, z)$. So

$$\Phi_{\text{new}} = \max(\varphi z + 1, \alpha z + \alpha) \leq \varphi z + \alpha.$$

Since $\varphi z = \alpha^2(\alpha z) \leq \alpha^2\Phi$, we get $\Phi_{\text{new}} \leq \alpha^2\Phi + \alpha$. Then $\alpha + 7 \leq 7\alpha^2$ gives the claim.

Fibonacci step ($d = 2, p = 3$). The new top two are $(z + z' + 1, z)$. So

$$\Phi_{\text{new}} \leq \max(\varphi z + z', \alpha(z + z')) + \alpha.$$

Now $\alpha(z + z') \leq \varphi z + z'$, because $(\varphi - \alpha)z \geq (\alpha - 1)z'$ follows from $z \geq z'$ and $\varphi - \alpha \approx 0.444 > \alpha - 1 \approx 0.174$. Also $\varphi z + z' = \varphi(z + z'/\varphi) \leq \varphi\Phi = \alpha^3\Phi$. Hence $\Phi_{\text{new}} \leq \alpha^3\Phi + \alpha$, and $\alpha + 7 \leq 7\varphi$ finishes.

Conclusion. Multiplying over all steps, $\Phi + 7 \leq 7\alpha^C$. Since every value is at most $z \leq \Phi$, we get $u < 7\alpha^C = 7\varphi^{C/3}$. $\square$

Remark. This proves that the golden ratio is the true speed limit, rigorously, for a natural class of libraries. It is short enough to formalize in Lean in a few hundred lines; the only nontrivial part is the dominance lemma, about multisets. It also explains the role of the *copy* piece αz . Without it, a copy step can multiply $z + z'/\varphi$ by φ at price 2, which is too much. With it, the copy step is paid for exactly ($\varphi = \alpha^2 \cdot \alpha$).

4 The obstruction: runs of near-equal values

Allowing three citations breaks this proof. Here is exactly why.

The bad state. Two copy steps from a value v produce $(v+2, v+1, v) \approx (v, v, v)$: three nearly equal top values. A window-3 step then gives about $3v$. The two-piece potential has $\Phi(v, v, v) = \varphi v$, and after the step $\Phi \approx (3 + 1/\varphi)v = (2 + \varphi)v \approx 3.618v$. But the allowed factor is $\alpha^4 \varphi \approx 3.07$. The potential is *undervaluing* the state (v, v, v) : from there, “window 3, then Fibonacci” really is worth about $1.905v$, not φv .

Patching creates more holes. We can add that strategy as a third piece, $\alpha^{-4}(\varphi z_1 + z_2 + z_3)$. Then other two-move openings matter: for example, “window 2, then window 3” is worth $\varphi^{-1/3}(z_1 + z_2)$ when $z_1 \approx z_2$. With m near-equal values, the best opening may be “sum six of them, copy, then Fibonacci”, and so on. The true value function, normalized by α^{cost} , is a supremum over infinitely many opening strategies.

What we know rigorously about this obstruction.

- **No linear potential works.** By linear-programming duality, no potential that is *linear* in the sorted values can certify any rate below 1.2198. This was proved earlier in the project.
- **Two natural relaxations overshoot.** Decoupled recursions, which bound each cited group by its own worst case, overshoot the truth. They give rate about 1.179 instead of 1.174, departing from the Lucas values from cost 33 on. (This is the single computation I ran for these notes, to test a proof strategy.)
- **Why they overshoot.** A relaxation lets “the result at position k ” play different roles for different citing results. In a real library it is *one* result. Any proof must keep this coupling.

Why this matters conceptually. Near-equal runs are expensive to build (2 symbols per copy) and cheap to exploit (one big window sums them all). A proof has to show that building a run never pays for itself. That is an amortization argument across many steps, not a one-step inequality, and it is the real content of the conjecture.

5 Ideas for a full proof

1. **Amortize over runs.** Group the recurrence into blocks: a run of copies followed by the window step that exploits it. Prove a block inequality: a block of price p multiplies a suitable potential by at most α^p .
Supporting evidence. The simplest block, “ $m-1$ copies then a window- m sum”, yields $\max(m + 1/\varphi, \alpha m)$ times the old maximum, at price $3m-1$. A short check shows this is at most φ^m , with equality exactly at $m=1, 2$. So the golden bound survives these blocks, tightly at the Fibonacci step. What remains is to control runs that are reused by later windows.
2. **A finite closed family of pieces.** Find finitely many “opening then Fibonacci” pieces whose maximum is preserved by every move, on *reachable* states only. Our earlier failed attempt used all sorted vectors. Reachable states are far more constrained: each value is exactly a window sum of earlier ones. Restricting to them may close the family.
3. **Symbolic dynamics and pressure.** The extremal number $2L_m - 2$ is twice the trace of $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^m$, minus 2. L_m counts the periodic orbits of period m of the *golden-mean shift*: binary

sequences on a cycle with no two adjacent 1s. This suggests the paths of an optimal library are a cyclic golden-mean code in disguise.

In that language, the conjecture is a *pressure inequality*: path entropy per unit of cost is at most $\frac{1}{3} \log \varphi$. The twist, and the reason standard thermodynamic formalism does not apply directly, is that cost is paid *once per result*, not once per visit along a path. That is precisely the difference between naming and copying.

4. **Entropy of a random path.** Take a uniformly random citation path. Its entropy is $\log u$, and it splits into local choices (stop, or follow one of d citations) weighted by visit probabilities. A Shearer-type inequality that charges each result’s entropy to its price $1 + d$, while using that visit probabilities decay geometrically along near-equal runs, could give the bound.

My judgment is that route 1 is the most likely to work, possibly combined with route 2. If it works, the result is a complete proof of the rate $\varphi^{1/3}$. The exact value $2L_m - 2$ would then need a further sharpening argument.

6 Frege vs. Extended Frege: what I can and cannot say

The dictionary. This is where our project genuinely helps understanding.

Our project	Circuits	Proofs
copying (unfolded size)	formulas	tree-like / Frege
naming (named size)	circuits	dag-like / Extended Frege
naming a <i>proved lemma</i>	sharing a gate output	reusing a line
naming a <i>new definition</i>	introducing a wire for a subcircuit	extension variable $q \leftrightarrow \varphi$

The trivial versus the deep direction. Our speed limit is a statement about *naive* unfolding: how much bigger a named object gets when every name is copied out. That direction is easy and elementary, and we proved it.

The deep direction asks whether there is *some other* small unnamed object doing the same job.

- For lemmas in Frege the answer is yes, up to a polynomial: Krajíček’s simulation of dag-like by tree-like Frege.
- For circuits versus formulas, and for definitions in proofs (Extended Frege versus Frege), nobody knows.

Our theorems do not touch the deep direction.

Why it is so hard.

- A lower bound must defeat *every* possible Frege proof, not one given proof. Our calculus always fixes the proof and only varies the naming.
- *Feasible interpolation*, one of the main tools for proof lower bounds, provably fails for Frege and for Extended Frege if standard cryptography (such as factoring) is hard. This is due to Krajíček–Pudlák and Bonet–Pitassi–Raz. So the method that works for weaker systems cannot work here.
- The circuit analogue, super-polynomial formula lower bounds, is stuck at roughly cubic. Researchers believe new ideas are needed; the Karchmer–Raz–Wigderson conjecture is one proposed route.

Could our ideas help at all? Two honest, small possibilities.

1. **The exact value $(o - 1)(W - 1) - 1$ and the rank-one (Schur) updates** describe precisely how much sharing a *given* circuit or proof benefits from. A lower-bound argument needs to show that *every* small circuit for some function has a bad sharing structure. Our calculus could be a bookkeeping tool inside such an argument. It is not the argument.
2. **Submodularity** (concepts are substitutes) says the benefit of sharing has diminishing returns within a fixed object. Separations, if they exist, must come from objects where sharing is *forced*, not just useful. Our bounded-reader result, where names become complements under capacity limits, hints at where to look: statements whose proofs cannot fit in bounded working memory without definitions. In proof complexity, a closely related precise notion is *proof space*. Whether a space-flavored argument could separate Extended Frege from Frege is, to my knowledge, not something anyone has made work.

7 Did our paper help? Summary

For the Lucas conjecture: yes, concretely.

- The calculus suggested thinking of a library as a multiset of values. That led to greedy dominance and the window form.
- The failed LP certificate told us that linear arguments cannot work, which pointed to the two-piece potential.
- That gave a real new theorem (Theorem 2): the golden speed limit is exact for binary libraries.
- We identified the precise obstruction for the general case.

For Frege vs. Extended Frege: only as a lens. The project clarifies what “naming versus copying” means and why the easy direction is easy. It offers no attack on the hard direction. This should be stated plainly in any write-up.

Suggested next step. Formalize Theorem 2 in Lean. Then try the block-amortization route for windows of size at most 3, the first case beyond binary. If that works, the general case may follow the same pattern.